SHIMADZU

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LabSolutions™ DB/CS

Instruction Manual

System Users Guide [LabSolutions TOC Volume]

Read this manual thoroughly before you use the product.
Keep this manual for future reference.

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Introduction

Read this Instruction Manual thoroughly before using the product.

Thank you for purchasing this product.

This manual describes the operation and options for this product. Read this manual thoroughly before using the product and operate the product in accordance with the instructions in this manual.

Keep this manual for future reference.

Important

- If the user or usage location changes, ensure that this manual is always kept together with the product.
- If this manual or a product warning label is lost or damaged, immediately contact your Shimadzu representative to request a replacement.
- To ensure safe operation, contact your Shimadzu representative if product installation, adjustment, re-installation (after the product is moved) is required.

Notice

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Original version is approved in English.

Indications Used in This Manual

The following symbols are used in this manual:

Indication	Meaning
NOTE	Emphasizes additional information that is provided to ensure the proper use of this product.
Reference	Indicates the location of related reference information.

Mouse and Window Operations

Description	Meaning
Click	Means pointing to an object and pressing the left mouse button once.
Right-click	Means pointing to an object and pressing the right mouse button once.
Double-click	Means pointing to an object and pressing the left mouse button twice.
Text in square brackets [ ]	The names of buttons in a window, and window names, are enclosed in square brackets. Example: Click [OK].
Text in quotation marks " "	Entered numerical values, text, and keyboard key names are indicated enclosed in quotation marks. Example: Press the "N" key while holding down the "Ctrl" key.

Use of Instruction Manuals

Shimadzu TOC analyzers include the LabSolutions TOC instrument control software and the LabSolutions Manager data management software as the LabSolutions system.
The instruction manuals, including this manual, are organized as follows.

Hint For details on how to operate the TOC analyzer, refer to the instruction manual for your particular model.

Instruction Manual	Description
LabSolutions Manager DB Installation & Maintenance Guide or LabSolutions Manager CS Installation & Maintenance Guide	This manual explains software installation, troubleshooting, and environment settings for the LabSolutions system.
LabSolutions DB/CS Installation & Maintenance Guide [LabSolutions TOC Volume]	This manual explains LabSolutions TOC model-specific installation, troubleshooting, and environment settings.
LabSolutions Manager DB System Users Guide or LabSolutions Manager CS System Users Guide	This manual explains how to use the LabSolutions system management function.
LabSolutions DB/CS System Users Guide [LabSolutions TOC Volume] (This manual)	This manual explains how to use LabSolutions TOC model-specific management functions.

For details on each LabSolutions command and execution method, refer to the provided Help file.

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Shimadzu provides the following warranty for this product.

1. **Period:**

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2. **Description:**

   If a product/part failure occurs for reasons attributable to Shimadzu during the warranty period, Shimadzu will repair or replace the product/part free of charge. However, in the case of products which are usually available on the market only for a short time, such as personal computers and their peripherals/parts, Shimadzu may not be able to provide identical replacement products.

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   (3) Product use in combination with hardware or software other than that designated by Shimadzu

   (4) Computer viruses leading to device failures and damage to data and software, including the product's basic software

   (5) Power failures, including power outages and sudden voltage drops, leading to device failures and damage to data and software, including the product's basic software

   (6) Turning OFF the product without following the proper shutdown procedure leading to device failures and damage to data and software, including the product's basic software

   (7) Reasons unrelated to the product itself

   (8) Product use in harsh environments, such as those subject to high temperatures or humidity levels, corrosive gases, or strong vibrations

   (9) Fires, earthquakes, or any other act of nature, contamination by radioactive or hazardous substances, or any other force majeure event, including wars, riots, and crimes

   (10) Product movement or transportation after installation

   (11) Consumables and equivalent items

   Recording media such as CD-ROMs are considered consumable items.

* If there is a document such as a warranty provided with the product, or there is a separate contract agreed upon that includes warranty conditions, the provisions of those documents shall apply.

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(2) This Agreement shall be construed and governed in accordance with the laws of Japan, excluding its conflict of law rules.

(3) The exclusive jurisdiction for any disputes arising out of or in connection with this Agreement shall be Kyoto District Court of Japan.

(4) The invalidity or unenforceability of any provision of this Agreement shall not affect the validity or enforceability of any other provision.

After-Sales Service and Availability of Replacement Parts

■ After-Sales Service

If any problem occurs with this product, perform an inspection and take appropriate corrective action as described in the "LabSolutions Manager DB (CS) Installation & Maintenance Guide", "LabSolutions DB/CS Installation & Maintenance Guide (LabSolutions TOC Volume)", and Help file.

If the problem persists, or the symptoms are not covered in the troubleshooting section, contact your Shimadzu representative.

■ Replacement Parts Availability

Replacement parts for this product will be available for a period of seven (7) years after the product is discontinued. Thereafter, such parts may cease to be available.

Note, however, that the availability of parts not manufactured by Shimadzu shall be determined by the relevant manufacturers.

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1 System Administration

This chapter explains the system administrator functions that are specific to the TOC model.

Reference For details on other functions, refer to the LabSolutions Manager CS or LabSolutions Manager DB System Users Guide.

1.1 System Administration Policy Settings

Reference For details, refer to "■ Instrument Policies" in "1.2.1 Set System Administration Policy (Security Policy)" in the LabSolutions Manager CS or LabSolutions Manager DB System Users Guide.

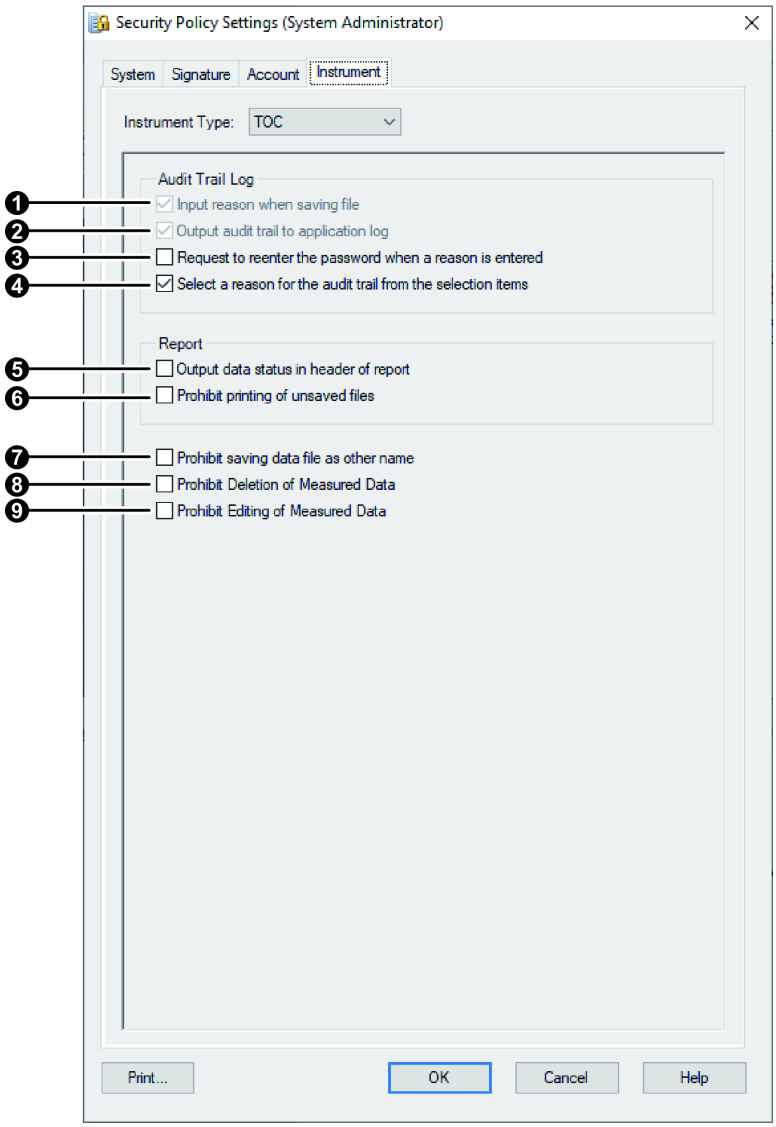
1.1.1 Instrument Policies

1 **Double-click the (Security Policy) icon in [Administration] of the [LabSolutions Main] window.**

2 **Click the [Instrument] tab.**

3 **Select [TOC] from the [Instrument Type] list.**

4 Set each item and click [OK].



No.	Name	Function	ER/ES
Audit Trail Log			
❶	Input reason when saving file	If an existing (already saved) data files and measurement parameters files are edited, a window to input the reason for editing is displayed when saving the file.	ON
❷	Output audit trail to application log	Enable the audit trail log recorded when the data files and measurement parameters files were saved to be displayed in the application log of the [Log Browser] window as well.	ON
❸	Request to reenter the password when a reason is entered	The password is required to enter the change reason for the audit trail.	-

No.	Name	Function	ER/ES
❹	Select a reason for the audit trail from the selection items	Select the reason from the preset selection item list when entering the change reason for the audit trail. Set the selection items in (Selection Administration) of [Administration Tools] on the [LabSolutions Main] window. Reference For details of the "Selection Administration", refer to LabSolutions Help.	-
Report			
❺	Output data status in header of report	Output the data file's information to a report when printing without saving the data file.	ON
❻	Prohibit printing of unsaved files	Prohibits printing of files being edited.	ON
Others			
❼	Prohibit saving data file as other name	Prohibits duplication of data files by disabling [Save as...] in the [File] menu.	-
❽	Prohibit deleting of measured data	Prohibits deletion of measured data in the sample table.	ON
❾	Prohibit re-editing of measured data	Prohibits re-editing of measured data in the sample table.	ON

1.1.2 Audit Trail

If [Input reason when saving file] of the instrument policy is enabled (checked), inputting the reason for changes is required when saving the edited data files and measurement parameters files.

For a data file, the input reason for change is recorded in the data history of the data file.

Change history of the measurement parameter files is also saved in the file and you can see it from the instrument control window.

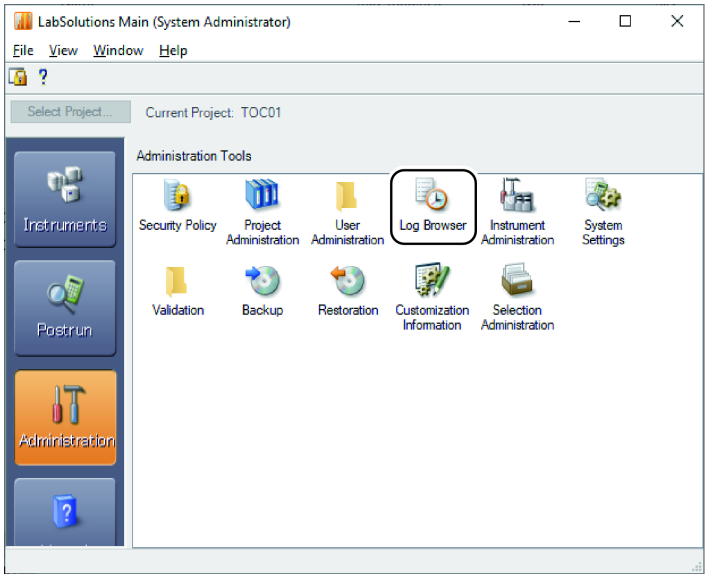
If [Output audit trail to application log] of the instrument policy is enabled, the changes and reasons can also be viewed in the [Log Browser].

This section explains about the audit trail log of a data file.

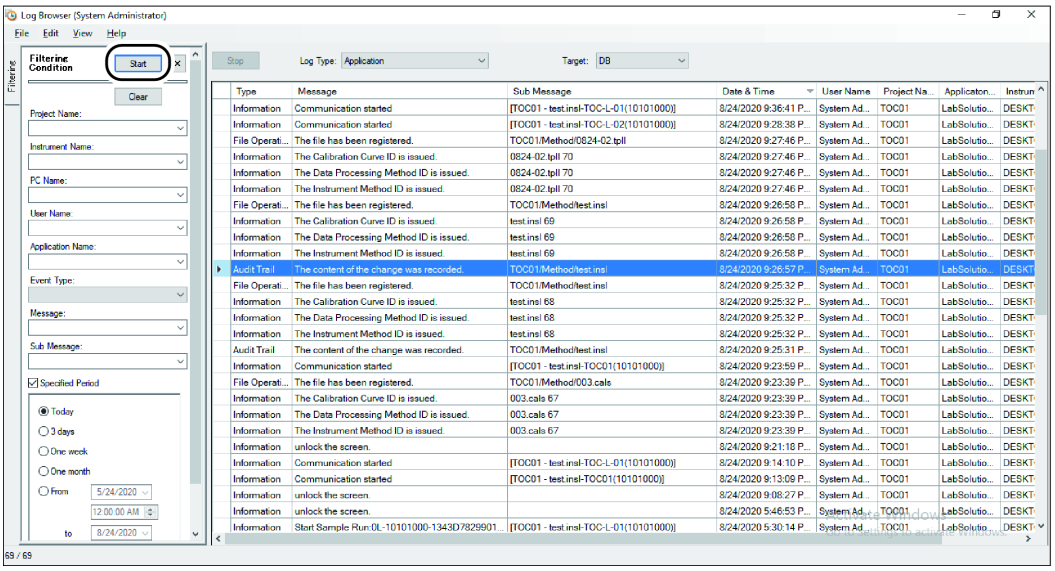
Reference For how to view the audit trail log of a measurement parameter file, see "3.3 Checking the Change History of Measurement Parameters" P.17.

■ Review Audit Trail Log Using the Log Browse

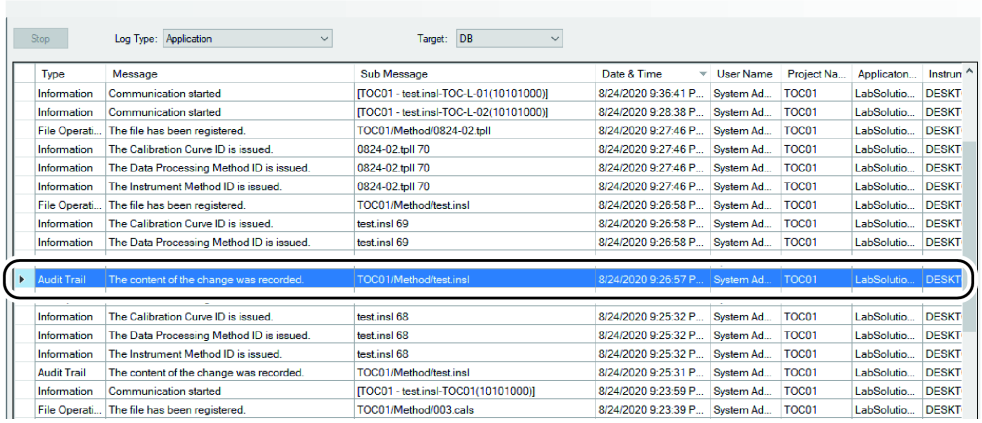
1 **Double-click the (Log Browse) icon in [Administration] of the [LabSolutions Main] window.**



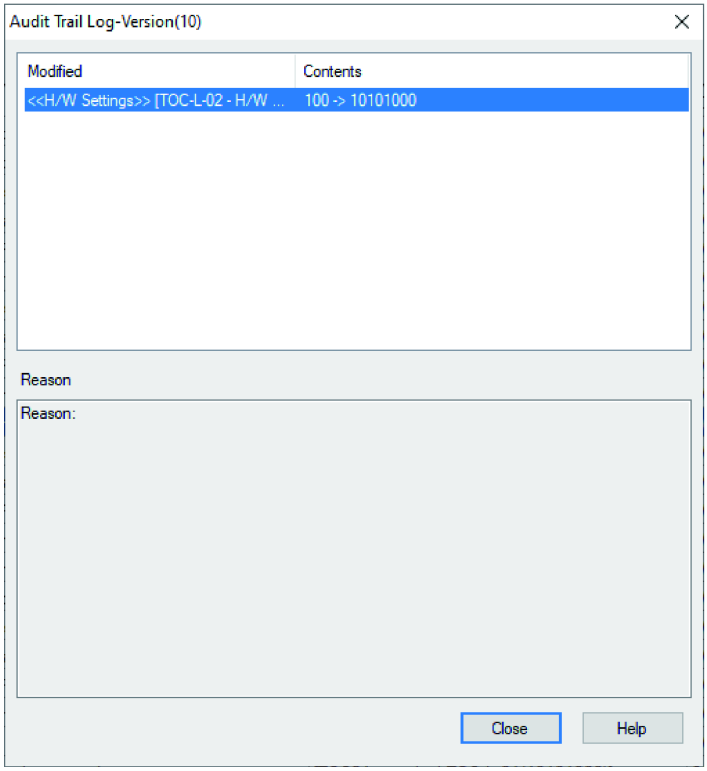
2 **Set filtering conditions and click [Start].**



3 Double-click the audit trail row.



The [Audit Trail Log] window is displayed.



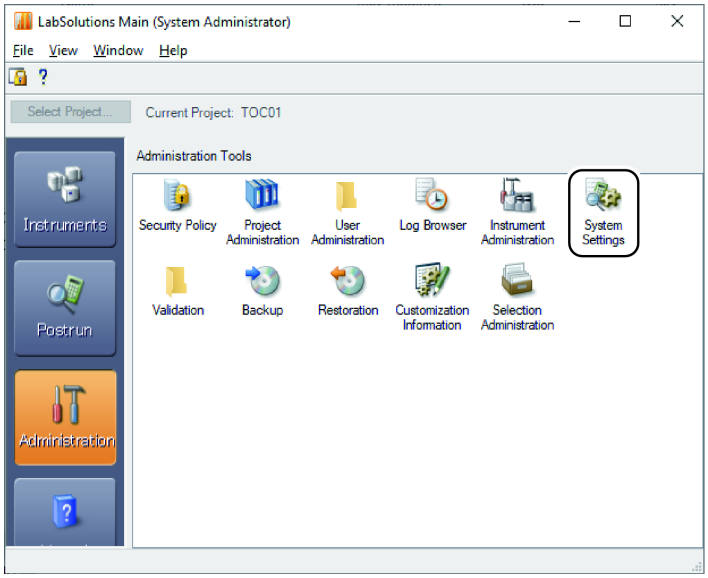
1.2 Numerical Rounding and Number of Displayed Digits

Reference For details, refer to "1.2.8 Numerical Rounding and Number of Displayed Digits" in the LabSolutions Manager CS or LabSolutions Manager DB System Users Guide.

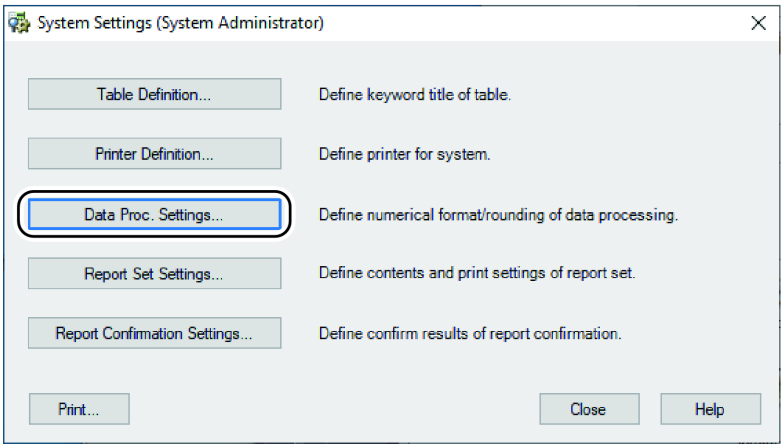
The software's [DataManager] rounding method and number of displayed digits can be batch-set in the [Data Processing Settings] sub-window.

NOTE Since system settings are changed in this sub-window, log in with a user ID having [Change System Settings] rights.

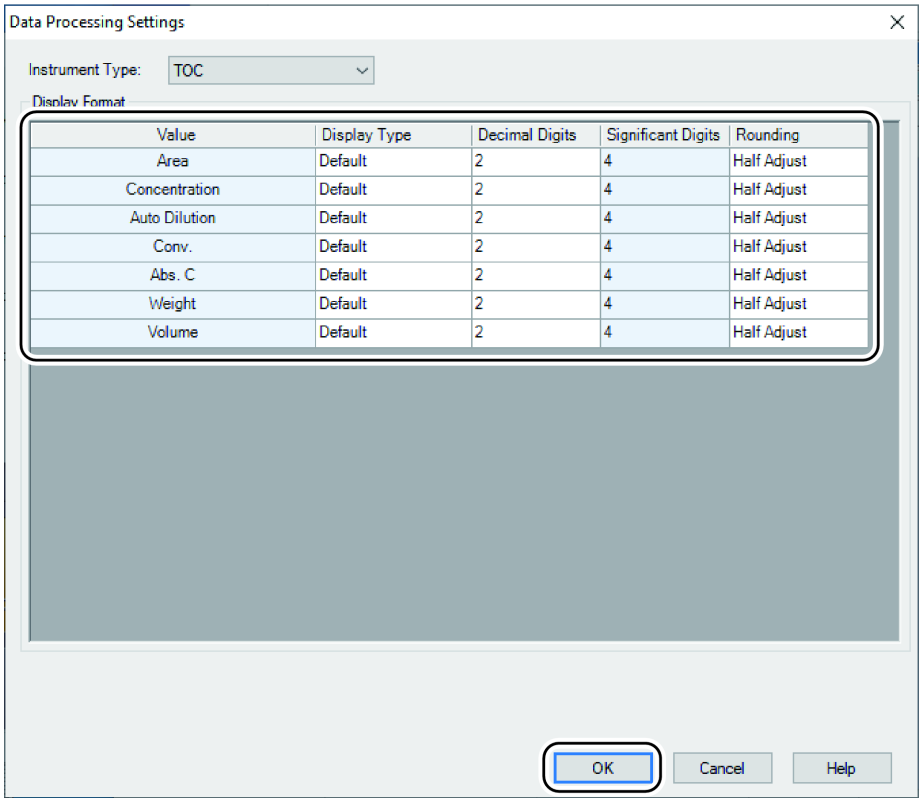
1 **Double-click the (System Settings) icon in [Administration] of the [LabSolutions Main] window.**



2 **Click [Data Proc. Settings].**



3 **Select the rounding method and number of displayed digits for specific calculation results, and click [OK].**



Value	Display Type	Decimal Digits	Significant Digits	Rounding
Area	Default	2	4	Half Adjust
Concentration	Default	2	4	Half Adjust
Auto Dilution	Default	2	4	Half Adjust
Conv.	Default	2	4	Half Adjust
Abs. C	Default	2	4	Half Adjust
Weight	Default	2	4	Half Adjust
Volume	Default	2	4	Half Adjust

NOTE If [Default] or [Exponential] is selected as the [Display Type], enter the number of decimal places in the [Decimal Digits] column, and if [Significant Digits] is selected, enter the number of digits in the [Significant Digits] column.

1.3 System Operation

Reference For details, refer to "1.3.2 Screen Lock to Protect System" in the LabSolutions Manager CS or LabSolutions Manager DB System Users Guide.

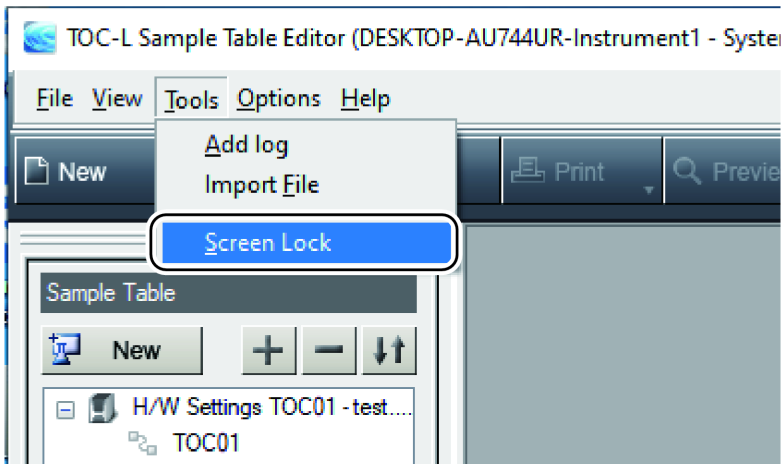
1.3.1 Screen Lock to Protect System

The windows of this software can be locked to protect the system from accidents or manipulation by other operators if the logged in operator temporarily leaves the PC.

When the lock function is enabled, the window is minimized and displayed as an icon on the taskbar. If the icon is clicked, the [User Authentication] window opens to prompt the user to enter the user ID and password.

This section describes an example of how to lock screens in the Spectrum measurement window.
The procedure for locking the TOC analyzer operation window is described here as an example.

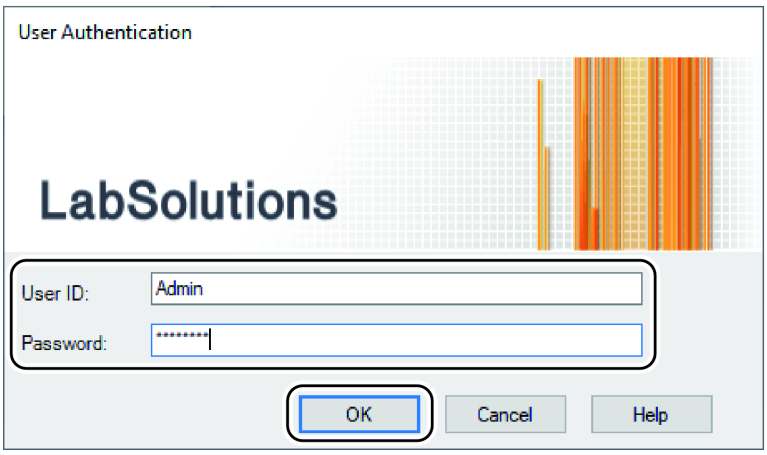
1 **Click [Screen Lock] in the [Tools] menu.**



The postrun window is minimized and an icon is displayed on the taskbar.

2 **Click the [Sample Table Editor] icon on the taskbar to open the application.**

3 **Enter a [User ID] and [Password], and click [OK].**



NOTE Only the user who locks the window can release the lock.

2 Data Management

This chapter explains the data administration functions that are specific to theTOC model.

For details on other functions, refer to the LabSolutions Manager CS or LabSolutions DB System Users Guide.

2.1 Data Management Using DataManager

[DataManager] allows for database information to be viewed for managing each file used by LabSolutions such as data files, method files, batch files, and report format files, and allows for the narrowing down of necessary data.

In addition, management is carried out while maintaining the links among the files. Therefore, it is easy to check the history and file version.

LabSolutions TOC files registered in [DataManager] are sorted according to the following types:

- Data
- Method

2.1.1 LabSolutions TOC File Formats

The following types of LabSolutions TOC files can be managed by [DataManager].

■ Data

Name	Ext.	Description
Data profile	dpll	This is the measurement result data file. The peak area value data, concentration data, measurement parameter information, file information, and data history are included.

Name	Ext.	Description
Sample table file	tlxl	This is the results table file used for measurement. The batch table information, measurement parameters, calibration curve parameters, file information, and data history are included.

NOTE

The data profile consists of the raw TOC data. The sample table file is a batch table file for performing measurement, though it can also be used to view the measurement result information.

The file version of sample table file is updated for each sample masurement.
So, it may be difficult to view the data profile when [History] is turned on in the DataManager during file version management.
In this case, specify "dpll" in [Data File Name] of [filtering] and filter.

If a communication failure occurs with the server when using the client server version, the application ends normally after the current sample measurement is completed.
In this case, the data profile of the sample that was being measured when the failure occurred is automatically registered in the database after communication with the server is restored.
However, it may not be possible for the sample table file to save the sample information that was being measured when the failure occurred.

■ Method

Name	Ext.	Description
System settings file	insl	The instrument settings, option settings, and change history are saved in this file.
Calibration curve settings file	cals	The concentration range, measurement parameters, and change history are saved in this file.
Calibration curve data file	cald	The concentration range, calibration curve, measurement parameters, and change history are saved in this file.
Method file	metl	The measurement parameters, calibration curve to use, and change history are saved in this file.
Control sample file	tpll	The measurement parameters, calibration curve to use, results check criteria, and change history are saved in this file.
Schedule file	tscl	This is the template file for the sample table file.

2.2 Other Functions in DataManager

2.2.1 Importing Local Files in the PC to the Database

Reference For details, refer to "2.2.1 Import Files on the PC to the Database" in the LabSolutions Manager CS or LabSolutions Manager DB System Users Guide.

With the LabSolutions [DataManager], there is a function for registering measurement parameters files and report files saved in local PC folders to the database.

NOTE It is possible to import data files, but data information cannot be extracted using the DataManager import function. Register data files to the database after loading them to LabSolutions TOC using the import function of LabSolutions TOC postrun application ([Import] from the [File] menu).

2.2.2 Data Set and Result File Operations

Reference For details, refer to "2.2 Other Functions in DataManager" and "■ Import Files Related to Data Acquisition Results as Result Files" in "2.2.3 Data Set and Result File Operations" in the LabSolutions Manager CS or LabSolutions Manager DB System Users Guide.

■ Import Files Related to Data Acquisition Results (Data Files) as Result Files

Files can be registered to the database related to a row (i.e. sample information) displayed in [Sample Information View] only when [Data] is selected as the file type.

NOTE
- To import a file, log in by a user ID having the [Import Files] rights.
- [DataManager] has a function for directly referencing the content of backup databases. Files, however, cannot be imported to backup database since backups are read-only.

2.2.3 Relating the File Type of Registered Data to Programs

Reference For details, refer to "2.2 Other Functions in DataManager" and "2.2.5 Relate the File Type of Registered Data to Programs" in the LabSolutions Manger CS or LabSolutions Manager DB System Users Guide.

With [DataManager], files contained in rows of sample information can be related to programs used for viewing and printing.

As the default, the following relations are set.

Name	Ext.	Description
Data profile	dpll, dpl	TocDataProfile.exe
Sample table file	tlxl	TOCSplTbl.exe
Calibration curve settings file	cals	TOCSplTbl.exe
Calibration curve data file	cald	TOCSplTbl.exe
Method file	metl	TOCSplTbl.exe
Control sample file	tpll	TOCSplTbl.exe

3 History Management

This chapter explains the operations for checking the change history of sample tables and measurement parameters created in LabSolutions TOC.

3.1 Creating History

LabSolutions TOC has the following functions for creating history.

Name	Description
Change history of sample table	This is the history that records all changes made to the sample table, from the time it was created until now.
Revision history of measurement parameters	This is the history that records all changes made to the measurement parameter file, such as changes to the method file, from the time it was created until now.
Change history of system settings file	This is the history that records all changes made to the system settings file for connecting to the instrument, from the time it was created until now.

The measurement results data creation history and the measurement start/stop operation history are recorded in the application log of the Log Browser.

Reference For details on the method for confirming history information from Log Browse, refer to "1.4.1 View History Information from the Log Browser" in the "LabSolutions Manager CS" or "LabSolutions Manager DB" System Users Guides.

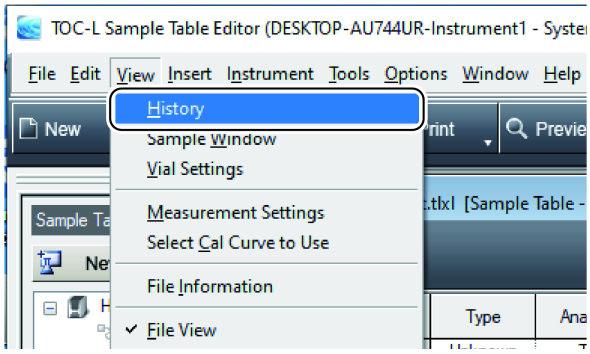
If [Input reason when saving file] of the instrument policy is enabled (checked), inputting the reason for changes is required when saving the edited data files and measurement parameters files.

Reference For how to input a reason for change, see "1.1.2 Audit Trail" P.3.

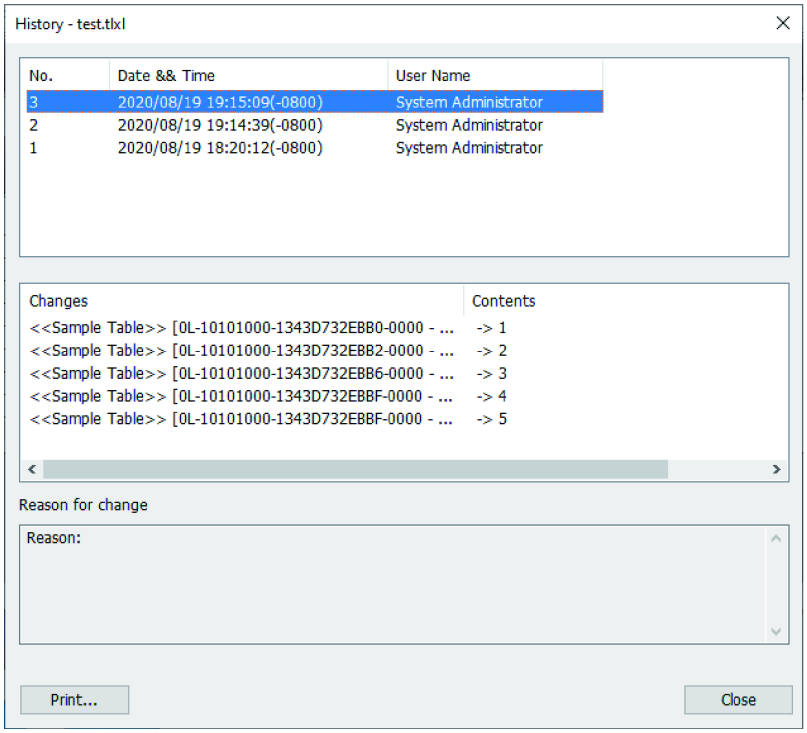
3.2 Checking the Change History of a Sample Table

View the change history that is saved in the sample table file by using the LabSolutions TOC measurement application.

1 **Click [View] - [History].**



The history of the current sample table is displayed.

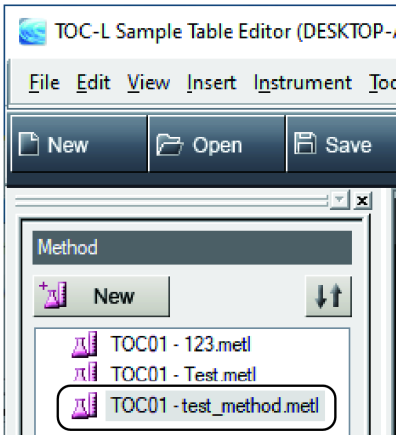


3.3 Checking the Change History of Measurement Parameters

3.3.1 Change History of a Measurement Parameter File in the Measurement Application

View the change history of a measurement parameter file by using the LabSolutions TOC measurement application.
This section describes an example of how to view the data history of the method file.

1 **Open the measurement parameter file.**

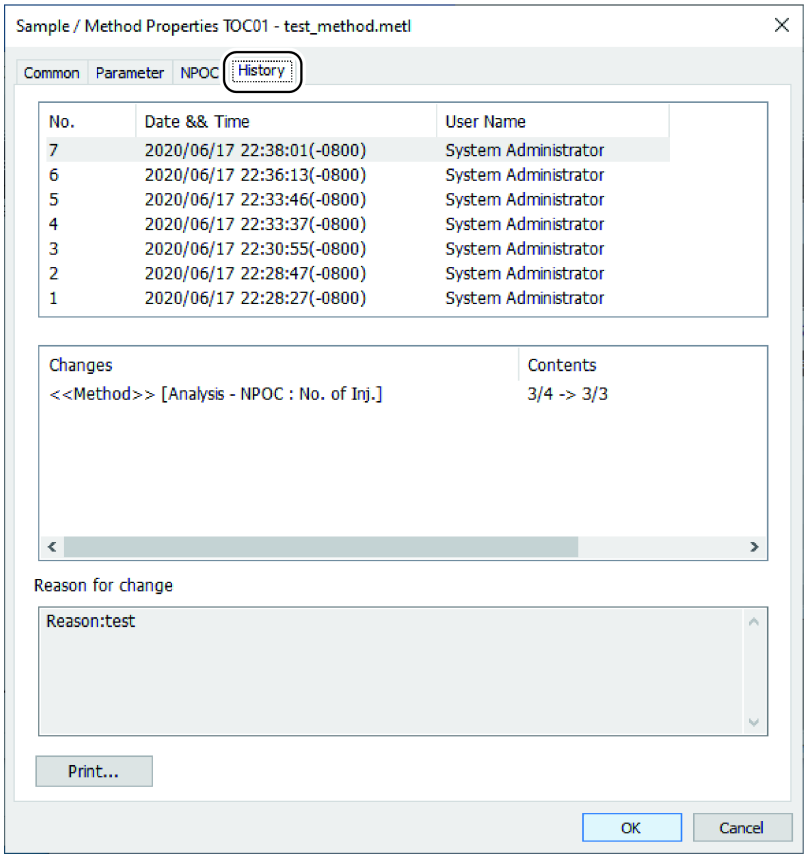


2 Click the [History] tab.

When clicking a desired history line (with the date, time and user name when the change is made) in the upper pain, the content of the change is displayed in the middle pane and the reason for the change in the lower pane.

Hint A reason for change can be input only when [Input reason when saving file] of the system administration policy is enabled.

Reference For the system administration policy, see "1.1 System Administration Policy Settings" P.1.

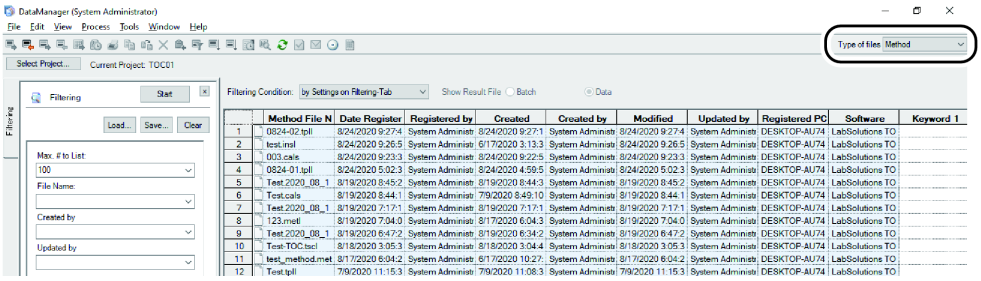


3.3.2 Change History of a Measurement Parameter File in a Project

View the change history of a measurement parameter file in a project by using DataManager.

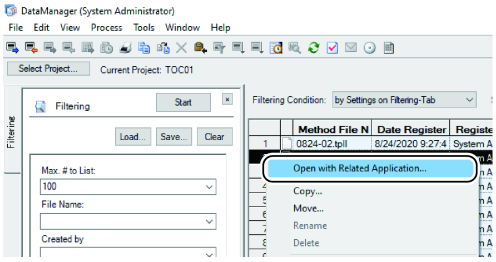
1 **Double-click the (DataManager) icon in [Administration] of the [LabSolutions Main] window.**

2 **In DataManager, select "Method" as the type of files.**

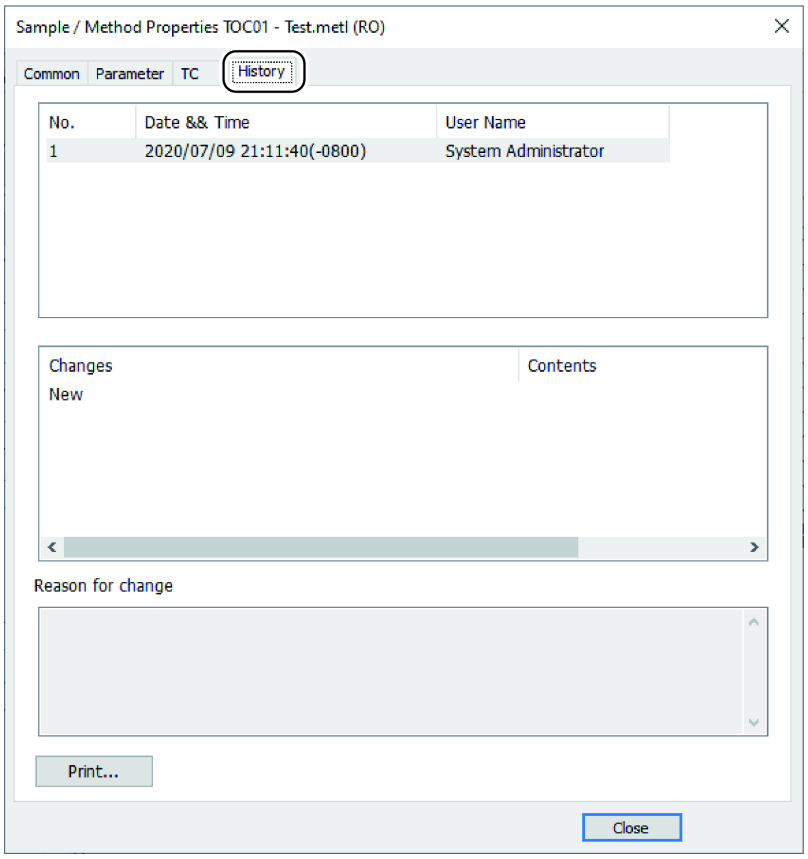


3

3 **Right-click a desired measurement parameter line and click [Open with Related Application] from the menu.**



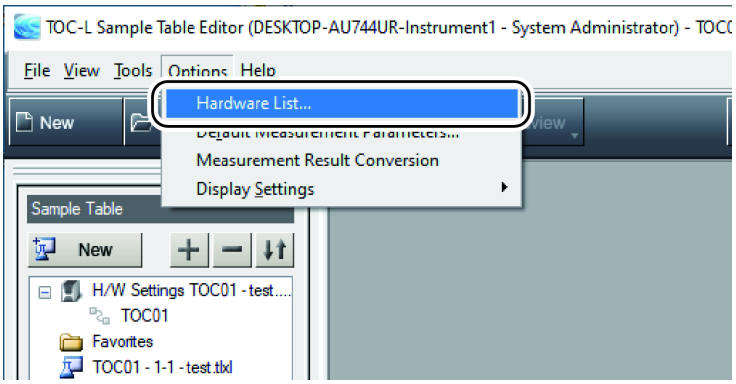
The details and change history of the selected measurement parameter are displayed.



3.4 Checking the Change History of the System Settings File

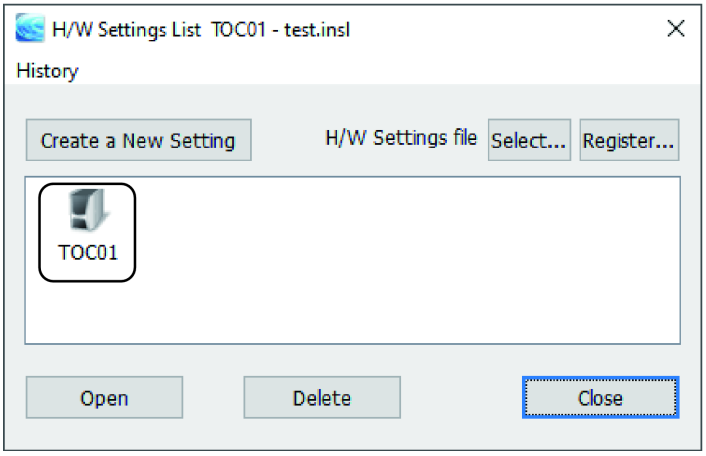
View the change history of the system settings file by using the LabSolutions TOC measurement application.

1 **Click [Opetion] - [Hardware List].**

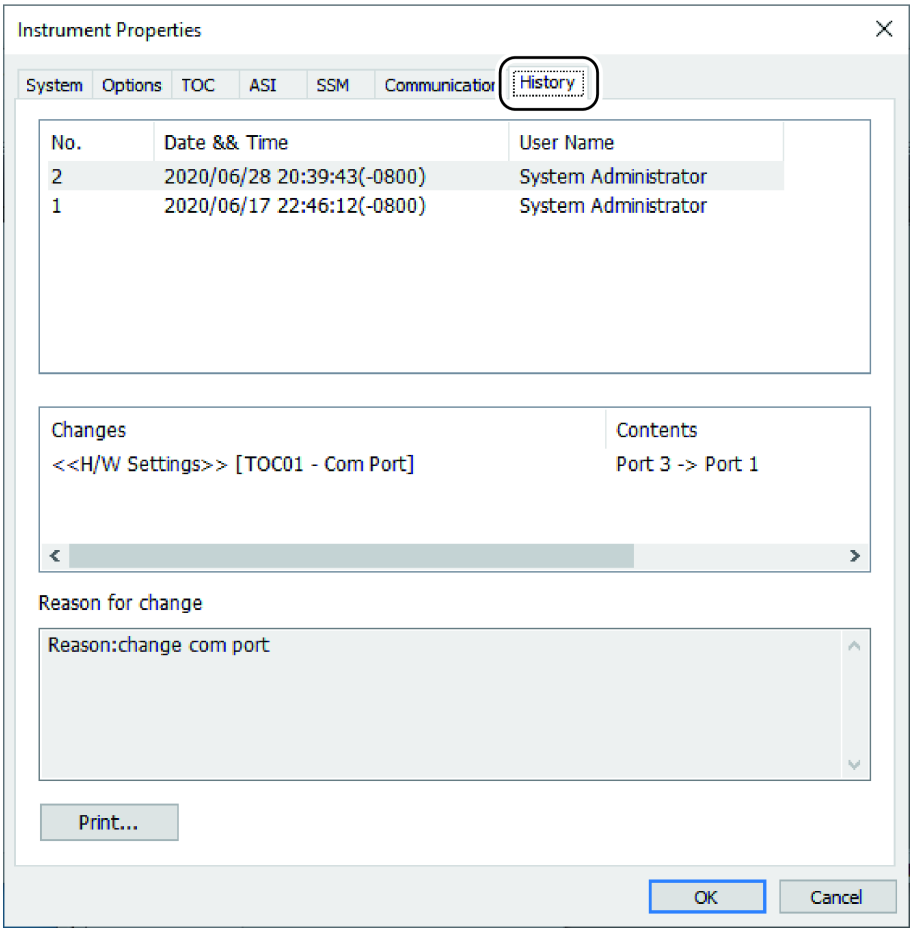


The [H/W Settings List] window opens.

2 **Click [TOC01].**



The [Instrument Propaties] window opens.

3 Click the [History] tab.

When you click a row in the history displayed in the upper box (the date and user name are shown for each change), the details of the change are shown in the middle box, and the reason for the change is shown in the lower box.